

Complete Math Roadmap

Canadian Curriculum + ML/CS Math — Aligned to Your Coding Timeline

Big Picture Alignment

Period	Coding Phase	Math Priority
Summer before Gr. 11	Phases 0-1 (DSA, Fundamentals)	MCR3U (Functions) + Linear Algebra start
Gr. 11 school year	Phases 0-2 (DSA, Backend)	MCR3U in school + Stats/Probability evenings
Summer before Gr. 12	Phases 2-4 (Backend, DevOps, System Design)	MHF4U (Advanced Functions) + finish Linear Algebra + Calculus entry
Gr. 12 school year	Phases 5-6 (System Design, Classical ML)	MHF4U + MCV4U in school + ML math consolidation evenings
Summer after Gr. 12	Phases 7-8 (Deep Learning, Applied AI)	Full ML math: Calculus, Probability, Information Theory, CS Math

Rule: Never let math block coding. Learn math just-in-time as your coding phases demand it. School carries the curriculum — use your own time for the CS-specific slices school won't teach.

How-to-Learn Shorthand

Label	What it means
Do-first	Attempt cold before reading or watching. Struggle is the point.
Watch → code same day	Watch 3B1B/StatQuest, then implement the concept in Python before the day ends.
Read → implement	Read one section, close it, implement the core idea from scratch. Never read two sections back-to-back without coding.
Timed reps	Do it under real time pressure with no hints open. Review only after a genuine attempt.

Label	What it means
Simulate, don't just calculate	Every stats/probability concept gets verified with a Python simulation the same day. <code>numpy.random</code> is your proof-checker.
Spaced re-solve	Re-derive or re-implement the same concept cold ~1 week later without notes.

☀ Summer Before Grade 11

~70 days · 1 hr/day math + your coding hours

Coding focus: finishing CS50P, starting C++, building the DSA habit. Math focus: MCR3U front-loaded so school feels like review, plus linear algebra now (no calculus needed).

Block 1 — MCR3U: Functions (Days 1–30)

1 hr/day. Finish before school starts so Gr. 11 math is revision, not new learning.

Days	Topic	Resource	Why It Matters for CS
1-3	Functions, domain/range, notation	Khan Academy — Functions	Function composition mirrors programming functions
4-6	Transformations (shift, reflect, stretch)	Khan Academy — Transformations	Activation function intuition later
7-8	Inverse functions	Khan Academy — Inverse Functions	Critical for bijections in crypto/hashing
9-11	Quadratics: vertex form, completing the square	Khan Academy — Quadratics	Optimization intuition
12-13	Quadratic formula, discriminant	Khan Academy — Quadratics	
14-15	Polynomial behaviour, factor/remainder theorem	Khan Academy — Polynomials	
16-17	Rational functions, asymptotes	Khan Academy — Rational Functions	
18-21	Exponential functions — growth/decay, transformations	Khan Academy — Exponential Functions	Direct prerequisite for learning rate schedules, sigmoid
22-23	Solving exponential equations	Khan Academy — Exponential Equations	
24-26	Trig — unit circle, radians, CAST rule	Khan Academy — Trigonometry	Fourier transforms later; positional encoding in transformers
27-28	Graphing sin/cos/tan, transformations	Khan Academy — Trig Graphs	
29-30	Sine law, cosine law	Khan Academy — Trigonometry	

Block 2 — Sequences & Series (Days 31-40)

Days	Topic	Resource	Why It Matters for CS
31-33	Arithmetic sequences & series	Khan Academy — Sequences	Algorithm complexity intuition
34-36	Geometric sequences & series	Khan Academy — Geometric Sequences	Infinite series, gradient descent convergence
37-38	Infinite geometric series	Khan Academy — Series	Neural net weight decay intuition
39-40	Pascal's triangle, intro to combinatorics	Khan Academy — Combinatorics	Probability foundation

Block 3 — Linear Algebra Start (Days 41-65)

This is Phase 5 of your coding roadmap pulled forward. No calculus needed.

Days	Topic	Resource	How to Learn
41-43	Vectors — addition, scalar mult., dot product	3Blue1Brown "Essence of Linear Algebra" Ep. 1-2	Watch → implement in NumPy same day
44-46	Linear transformations, matrix multiplication	3B1B Ep. 3-4	Watch → build matrix multiply by hand, verify with <code>np.dot</code>
47-49	Determinants	3B1B Ep. 5-7	Watch → implement
50-52	Systems of equations, Gaussian elimination	Khan Academy Linear Algebra	Read → solve a 3×3 system in pure Python
53-55	Matrix inverse, rank, null space	3B1B + Khan Academy	Watch/read → implement
56-58	Eigenvectors & eigenvalues	3B1B Ep. 14	Watch → find eigenvalues of a 2×2 by hand, verify with <code>np.linalg.eig</code>
59-61	PCA intuition (eigenvectors applied)	StatQuest "PCA" video	Watch → PCA on a toy 2D dataset in NumPy
62-64	Matrix norms, orthogonality, projections	Khan Academy	Read → implement
65	Review & consolidation	—	Build a mini linear algebra library in Python from scratch

Block 4 — Review & Buffer (Days 66-70)

Catch-up days. Re-solve anything shaky. No new material.

Grade 11 School Year

~10 months · ~30-45 min/day on math outside school

School handles MCR3U. Evening time goes to probability & statistics — the other half of Phase 5 that school won't cover adequately.

School — MCR3U (JensenMath)

Follow your teacher's pace. Use jensenmath.ca for video support on every unit — it covers the course exactly.

Unit	Topic	Resource	What's Covered
Unit 1	Functions	jensenmath.ca → Grade 11 → Unit 1	Domain & range, function notation, max/min quadratics, radicals, solving quadratics
Unit 2	Rational Expressions	jensenmath.ca → Grade 11 → Unit 2	Exponent laws, rational exponents, multiply/divide/add/subtract, solving quadratics
Unit 3	Transformations	jensenmath.ca → Grade 11 → Unit 3	Intro to transformations, transform x^2 , root x , $1/x$, inverse functions
Unit 4	Exponential Functions	jensenmath.ca → Grade 11 → Unit 4	Exponential growth/decay, interest, graphing, transformations
Unit 5	Trig Geometry	jensenmath.ca → Grade 11 → Unit 5	Special angles, ratios $>90^\circ$, solving trig equations, reciprocal ratios, ambiguous case of sine, trig identities
Unit 6	Trig Functions	jensenmath.ca → Grade 11 → Unit 6	Graphing sine and cosine, transformations, trig applications
Unit 7	Discrete Functions	jensenmath.ca → Grade 11 → Unit 7	Sequences, series, recursive functions, Pascal's triangle
Exam	Full Course Review	jensenmath.ca → Grade 11 → Full Course Review	Use the full review section before your exam

You already front-loaded all of this in the summer. School is consolidation and exam accuracy, not new learning.

Evenings — Probability & Statistics

~3 sessions/week, 30–45 min. Roughly October → March.

By the time you hit Phase 6 (Classical ML) in coding, you'll already understand why cross-entropy is the loss function and what MLE is actually doing.

Week	Topic	Resource	How to Learn
1-2	Counting: permutations, combinations	Khan Academy Stats	Timed reps on exercises
3-4	Basic probability: sample spaces, events, complement rule	StatQuest + Khan Academy	Watch → simulate in Python (flip coins, roll dice)
5-6	Conditional probability, independence	StatQuest	Watch → verify with Python simulations
7-8	Bayes' theorem	StatQuest "Bayes' theorem"	Watch → write a Naive Bayes classifier from scratch
9-10	Discrete distributions: Bernoulli, Binomial, Poisson	StatQuest	Watch → simulate each distribution in NumPy
11-12	Continuous distributions: Uniform, Normal (Gaussian)	StatQuest	Watch → sample, plot histograms, compute z-scores
13-14	Expectation, variance, standard deviation	Khan Academy	Read → implement by hand on small datasets
15-16	Central Limit Theorem	StatQuest	Watch → simulate: show CLT empirically in Python
17-18	Covariance, correlation	StatQuest	Watch → correlation matrix in NumPy
19-20	Maximum Likelihood Estimation (MLE)	StatQuest	Watch → MLE for a Gaussian by hand
21-22	Hypothesis testing, p-values (conceptual)	StatQuest	Watch → implement a t-test manually
23-24	Information theory basics: entropy, cross-entropy, KL divergence	Colah's blog	Read → compute cross-entropy loss manually

Summer Before Grade 12

~70 days · 1 hr/day

Coding focus: wrapping up Backend (Phase 2), DevOps (Phase 4), System Design (Phase 3). Math focus: MHF4U preview + calculus entry point.

Block 1 — MHF4U: Advanced Functions Preview (Days 1-25)

Front-load so Gr. 12 school math is easy.

Days	Topic	Resource
1-3	Polynomial functions — graphs, end behaviour, multiplicity of roots	Khan Academy — Polynomials
4-6	Dividing polynomials, factor theorem review	Khan Academy — Polynomials
7-9	Rational functions — intercepts, asymptotes (vertical, horizontal, oblique)	Khan Academy — Rational Functions
10-12	Logarithms — definition, laws, change of base	Khan Academy — Logarithms
13-15	Solving log equations, graphing log functions	Khan Academy — Logarithms
16-18	Exponential & log relationship (inverses)	Khan Academy
19-21	Trig identities — Pythagorean, sum/difference formulas	Khan Academy — Trig Identities
22-24	Solving trig equations, general solutions	Khan Academy — Trig Equations
25	Review	—

Block 2 — Calculus Entry Point (Days 26-55)

MCV4U preview. The chain rule IS backpropagation — go slow, master it.

Days	Topic	Resource	Why It Matters for CS
26-28	Limits — intuition, limit laws	Khan Academy — Limits	Foundation for derivatives
29-32	Derivatives from first principles	Khan Academy — Derivatives	Understanding what a gradient is
33-35	Differentiation rules: power, product, quotient	Khan Academy — Differentiation Rules	
36-39	Chain rule	Khan Academy — Chain Rule	This IS backpropagation. Go slow. Master it.
40-42	Derivatives of exponentials & logs (e^x , $\ln x$)	Khan Academy	Softmax, log-loss derivatives
43-45	Derivatives of trig functions	Khan Academy	
46-48	Partial derivatives — intro	Khan Academy — Multivariable	Direct prerequisite for gradient descent in ML
49-51	Gradients & gradient vectors	Khan Academy — Multivariable	The math behind every optimizer (SGD, Adam)
52-54	Optimization — critical points, 2nd derivative test	Khan Academy — Optimization	Loss minimization intuition
55	Review	—	

Block 3 — Linear Algebra Completion + Multivariable (Days 56-70)

Days	Topic	Resource	Why It Matters for CS
56-58	Jacobian matrix intuition	MML Book Ch. 5	Backprop through layers
59-61	Gradient descent implemented in NumPy	—	Linear regression with gradient descent from scratch — no sklearn
62-64	SVD intuition (Singular Value Decomposition)	StatQuest "SVD"	Used in recommendation systems, compression

Days	Topic	Resource	Why It Matters for CS
65-67	Vector calculus: gradient, divergence (light)	MML Book	Needed for understanding loss landscape
68-70	Backpropagation from scratch — derive it mathematically	Karpathy "Micrograd"	Implement a 2-layer net backward pass in pure NumPy

Grade 12 School Year

~10 months

School handles MHF4U (Advanced Functions) and MCV4U (Calculus & Vectors). Evening time goes to solidifying ML math and beginning deeper CS math.

School — MHF4U Advanced Functions (JensenMath)

You've already previewed this entire course. School is consolidation and exam accuracy.

Unit	Topic	Resource	What's Covered
Unit 1	Polynomial Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 1	Graphs, end behaviour, multiplicity — fully previewed
Unit 2	Rational Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 2	All asymptote types — fully previewed
Unit 3	Exponential & Logarithmic Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 3	Fully previewed — focus on exam accuracy
Unit 4	Trigonometric Functions & Identities	jensenmath.ca → Grade 12 → MHF4U → Unit 4	Identities, equations, general solutions — fully previewed
Unit 5	Rates of Change	jensenmath.ca → Grade 12 → MHF4U → Unit 5	Bridge to MCV4U — limits and intro derivative concepts
Exam	Full Course Review	jensenmath.ca → Grade 12 → MHF4U → Full Course Review	

School — MCV4U Calculus & Vectors (JensenMath)

You've previewed derivatives and the chain rule. School fills in the full picture.

Unit	Topic	Resource	What's Covered
Unit 1	Derivative Rules	jensenmath.ca → Grade 12 → MCV4U → Unit 1	Power, product, quotient, chain rule, applications — previewed, now precision-drill for exams
Unit 2	Curve Sketching	jensenmath.ca → Grade 12 → MCV4U → Unit 2	Increasing/decreasing, concavity, 1st & 2nd derivative tests, optimization
Unit 3	Derivative of Trig & Exponentials	jensenmath.ca → Grade 12 → MCV4U → Unit 3	Implicit differentiation, applications — e^x and $\ln x$ derivatives connect to ML loss functions
Unit 4	Geometric Vectors	jensenmath.ca → Grade 12 → MCV4U → Unit 4	Vector addition, scalar multiplication, force/velocity/tension — physical intuition for ML vectors
Unit 5	Algebraic Vectors	jensenmath.ca → Grade 12 → MCV4U → Unit 5	Dot product, cross product, vectors in 3-space — connects directly to your linear algebra work
Unit 6	Lines & Planes	jensenmath.ca → Grade 12 → MCV4U → Unit 6	Vector and scalar equations, intersecting lines and planes in 3-space
Exam	Full Course Review	jensenmath.ca → Grade 12 → MCV4U → Full Course Review	

Evenings — ML Math Consolidation

~3 sessions/week, 30–45 min. Mathematics for Machine Learning (free at mml-book.github.io) is your capstone text. Read it now that school has given you the calculus foundation.

Period	Topic	Resource
Sept-Oct	Multivariate calculus — chain rule in matrix form, Jacobians	MML Book Ch. 5
Nov-Dec	Optimization theory — convexity, saddle points, Adam/momentum math	MML Book Ch. 7
Jan-Feb	Linear algebra deep dive — SVD, matrix factorization, spectral methods	MML Book Ch. 2-4
Mar-Apr	Probability theory formal — measure theory lite, conjugate priors, distributions in depth	MML Book Ch. 6
May-Jun	Information theory — entropy, mutual information, KL divergence applications	Elements of Information Theory Ch. 1-2

☀ Summer After Grade 12

~70 days — aligns with Phases 7-8: Deep Learning + Applied AI

This is where the math pays off: implementing backprop, attention mechanisms, and loss functions for real.

Block 1 — Deep Learning Math (Days 1-30)

Days	Topic	Resource	Coding Tie-in
1-5	Backprop derivation — scalar $\rightarrow$ vector $\rightarrow$ matrix form	Karpathy "Micrograd" + MML Ch. 5	Phase 7: implement in pure NumPy
6-10	Loss functions derived — MSE, cross-entropy, KL divergence	MML Book	Why these functions, not others
11-15	Activation functions — sigmoid, tanh, ReLU, derivatives	MML Book + d2l.ai	Vanishing gradient problem — understand it mathematically
16-20	Attention mechanism math — scaled dot-product, softmax derivation	3B1B "Attention" + d2l.ai	Phase 7: implement attention from scratch in PyTorch

Days	Topic	Resource	Coding Tie-in
21-25	Convolutional math — cross-correlation, filter math	d2l.ai CNN chapter	CNN intuition
26-30	Regularization math — L1/L2 as priors (Bayesian view), dropout	MML Book Ch. 8	Why regularization works, not just how

Block 2 — Advanced CS Math (Days 31-55)

Days	Topic	Resource	Why It Matters
31-35	Combinatorics & counting — generating functions, recurrences	Competitive Programmer's Handbook	DSA: DP state counting, CCC problems
36-40	Graph theory — trees, cycles, planarity, graph representations	USACO Guide Gold + CP Handbook	DSA: Phase 0 Gold topics
41-45	Discrete math — logic, sets, relations, proofs	CP Handbook	CS fundamentals, algorithm correctness
46-50	Number theory — modular arithmetic, Fermat's little theorem, RSA	CP Handbook — number theory chapters	Cryptography in backend/security
51-55	Big-O formal — asymptotic notation proofs, recurrence relations (Master theorem)	CP Handbook + CLRS reference	Analyzing your own algorithms rigorously

Block 3 — Bayesian & Statistical Learning Theory (Days 56-70)

Days	Topic	Resource	Coding Tie-in
56-59	Bayesian ML — Bayesian linear regression, MAP estimation	MML Book Ch. 9	Probabilistic models in Phase 8
60-63	Statistical learning theory — bias-variance tradeoff formally, VC dimension (intro)	MML Book + ESL reference	Understanding why your models generalize
64-67	Embedding math — vector spaces, cosine similarity derivation, why	MML Book + Phase 4 embedding work	Grounding your RAG and embedding work in math

Days	Topic	Resource	Coding Tie-in
	dot product works		
68-70	Review + fill gaps from Phase 7-8 coding work	—	Identify what actual coding exposed as weak — patch those

Complete Curriculum Checklist

Ontario School Curriculum (all covered)

- ☐ MPM1D/2D (Gr. 9-10) — assumed done
- ☐ MCR3U (Gr. 11 Functions) — Summer before Gr. 11 + school year
- ☐ MHF4U (Gr. 12 Advanced Functions) — Summer before Gr. 12 + school year
- ☐ MCV4U (Gr. 12 Calculus & Vectors) — Summer before Gr. 12 + school year
- ☐ MDM4U (Data Management — optional) — covered by stats self-study, better than the course

ML/CS Math (all covered)

- ☐ Linear Algebra — Summer Gr. 11, finished Gr. 12 evenings
- ☐ Probability & Statistics — Gr. 11 evenings
- ☐ Calculus (derivatives, chain rule, partial derivatives) — Summer Gr. 12
- ☐ Multivariable Calculus / Gradient Descent math — Summer Gr. 12
- ☐ Information Theory — Gr. 12 evenings
- ☐ Backprop math — Summer after Gr. 12
- ☐ Attention math — Summer after Gr. 12
- ☐ Discrete Math / Combinatorics — Summer after Gr. 12
- ☐ Graph Theory — Summer after Gr. 12
- ☐ Number Theory — Summer after Gr. 12

Golden Rules

1. School carries the curriculum. Never spend self-study time re-doing what school covers adequately. Use your hours for the CS math school skips.

2. Linear algebra before calculus. You can start linear algebra now — no calculus needed. It pays off in Phase 6 before you even touch derivatives.

3. Simulate, don't just calculate. Every stats and probability concept should be verified with a Python simulation the same day. `numpy.random` is your proof-checker.

4. The chain rule is backprop. Spend more time here than anything else in calculus. Master it symbolically, then implement it in code.

5. MML book is your capstone math text. Mathematics for Machine Learning (free at mml-book.github.io) ties all of this together. Read it in Gr. 12 once school has given you the calculus foundation.